

Erythema Nodosum

Erythema nodosum is characterized by a sudden onset of symmetric, tender, erythematous, warm nodules and raised plaques with bilateral distribution, typically located on the anterior aspect of the lower extremities (Figs. 68-4 and 68-5). The lesions are most commonly found on the shins. Initially, the nodules appear vivid red or purplish, evolving over several days to a yellow or greenish hue, resembling a deep bruise—a feature known as *erythema contusiforme*. Ulceration does not occur, and healing takes place without atrophy or scarring.

Acute episodes of erythema nodosum are often accompanied by systemic symptoms such as fever, fatigue, malaise, arthralgia, headache, abdominal pain, vomiting, cough, or diarrhea. The eruption typically lasts 3 to 6 weeks, and recurrences are common.

Clinical variants such as erythema nodosum migrans, subacute nodular migratory panniculitis of Vilanova and Pinol, and chronic erythema nodosum are now considered part of the spectrum of erythema nodosum rather than distinct entities.

Etiology

Erythema nodosum is associated with a wide range of underlying conditions, including:

- **Infections** (e.g., streptococcal infections—most common in children; also tuberculosis, Yersinia, Mycoplasma, fungal infections)
- **Medications** (e.g., sulfonamides, oral contraceptives, penicillins)
- **Malignancies** (e.g., lymphoma, leukemia)
- **Autoimmune and inflammatory diseases** (e.g., sarcoidosis, inflammatory bowel disease—most common in adults)
- **Pregnancy**
- **Idiopathic** (a significant number of cases have no identifiable cause)

It is regarded as a hypersensitivity reaction to various inciting agents.

Histopathology

Erythema nodosum is the classic example of **mostly septal panniculitis without vasculitis**. Key histologic features include:

- Thickened connective tissue septa of the subcutis infiltrated with inflammatory cells extending into periseptal fat.
- Inflammatory infiltrate composition varies with lesion age:
- **Early lesions:** Predominance of neutrophils.
- **Late lesions:** Histiocytes, small granulomas, and multinucleated giant cells.
- **Miescher radial granuloma:** A characteristic finding consisting of small, round aggregates of histiocytes arranged radially around a central slit-like cleft. The nature of this cleft remains unknown, with no evidence of endothelial lining on ultrastructural or immunohistochemical studies.
- In older lesions, histiocytes may coalesce into multinucleated giant cells that retain a stellate cleft reminiscent of the Miescher granuloma.
- **Absence of vasculitis** is a critical diagnostic feature.
- Late-stage changes include granulation tissue at the septal-fat interface and fibrosis, but resolution occurs without scarring or atrophy.

Treatment

Management focuses on identifying and treating the underlying condition.

- Most cases resolve spontaneously within a few weeks.
- Supportive measures such as bed rest may be sufficient.
- Symptomatic relief can be achieved with:
- **Aspirin**
- **Nonsteroidal anti-inflammatory drugs (NSAIDs)**
- **Saturated solution of potassium iodide** (2–10 drops daily), which may accelerate resolution

- **Systemic corticosteroids** are rarely necessary and should be avoided unless absolutely indicated, particularly if an underlying infection has not been ruled out.

Mostly Lobular Panniculitis with Vasculitis

Erythema Nodosum Leprosum (See Chap. [Chapter reference removed])